

LC 1019: Next Greater Node in Linked List

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Approach 1: Brute Force (Time: $O(n^2)$)

The brute force approach suggests first converting the Linked List to an array to easily traverse and compare elements.

(2) → (1) → (5)

Output Array: [5, 5, 0]

- Let the converted array be **nums**.
- Let **ans** be the array we return, initialized entirely to 0.

```
for (int i = 0; i < n; i++) {
    for (int j = i + 1; j < n; j++) {
        if (nums[j] > nums[i]) {
            ans[i] = nums[j]; // we just need next greater
            break;
        }
    }
}
return ans;
```

Approach 2: Monotonic Stack (Time: $O(n)$)

We can optimize the lookup to linear time by maintaining a monotonic stack.

- **Step 1:** Create a `nums` array where we will store the linked list values.
- **Step 2:** Create an `ans` array where we will store answers, and initialize it to 0.
- **Step 3:** Use a Stack `st` to store the indices whose answers are not yet found.

```
stack<int> st; // store indices whose answers not yet found

for (int i = 0; i < n; i++) {
    while (!st.empty() && nums[st.top()] < nums[i]) {
        ans[st.top()] = nums[i];
        st.pop();
    }
    st.push(i);
}
return ans;
```